AWS NETWORKING PROJECT

VPC Peering: Private Connectivity Across Two VPCs

AWS Architect Associate • EWB Courses

Student name	_____
Batch	_____
AWS Region	Asia Pacific (Mumbai) — ap-south-1
Completion date	24 October 2026 (update if needed)
Project status	Core private connectivity verified; remaining evidence/tests noted in report

Project objective

Build a secure private connection between two non-overlapping Amazon VPCs using VPC Peering, route tables, scoped security group rules and a public bastion host for administration. Demonstrate reachability between private EC2 instances using private IPv4 addresses.

Report note: This draft uses the resource details and terminal results confirmed during the lab. The initial pre-peering baseline was not captured, and the controlled ICMP failure/recovery and cleanup evidence are still pending. Do not present those items as completed until you run them and add your own screenshots.

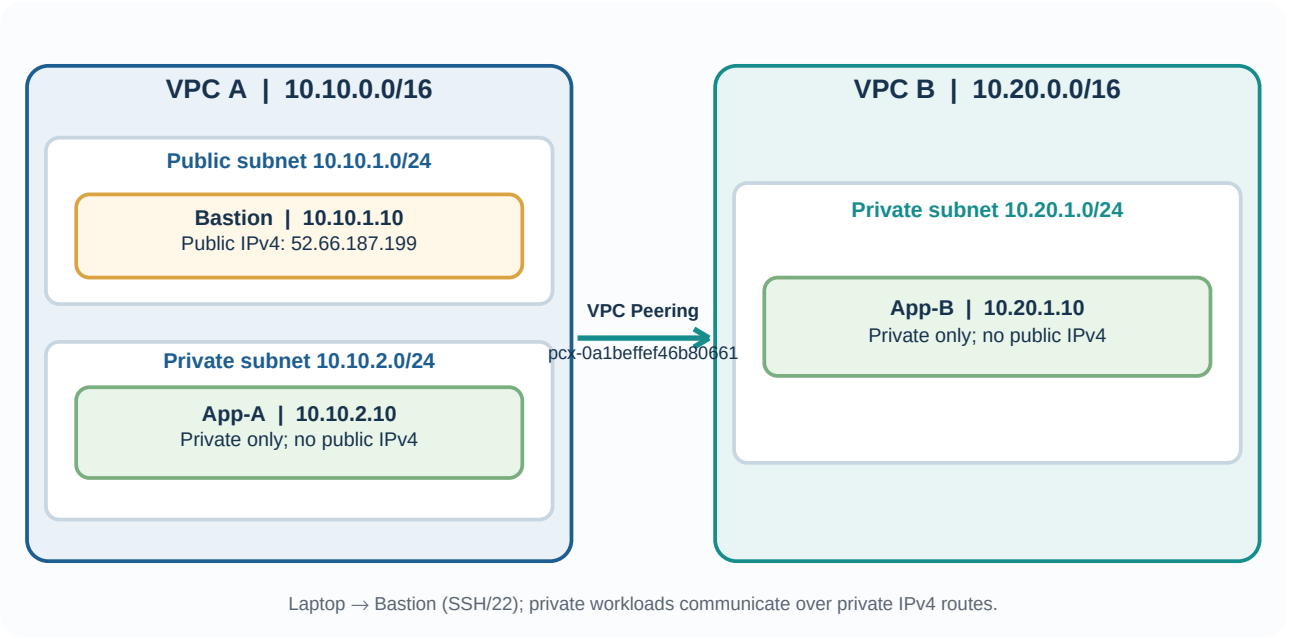
1. Executive Summary

This project establishes private IPv4 connectivity between VPC A (10.10.0.0/16) and VPC B (10.20.0.0/16) in the AWS Mumbai Region. A VPC peering connection links the VPCs. Explicit routes on both private route tables provide the forward and return paths. A route in VPC A's public route table also allows the bastion host to reach App-B over the peering connection for SSH administration.

The live tests completed so far confirmed successful ICMP communication in both directions: App-A (10.10.2.10) to App-B (10.20.1.10), and App-B back to App-A. Each test returned four replies with 0% packet loss. SSH access was also established from the laptop to the bastion, and through the bastion to each private instance.

The two application instances have no public IPv4 addresses. VPC B has no Internet Gateway; VPC peering provides private routing between the VPCs but does not give VPC B internet access through VPC A.

2. Architecture



Laptop → Bastion (SSH/22); private workloads communicate over private IPv4 routes.

Traffic paths

Administration: Laptop → public IPv4 of bastion (SSH/TCP 22) → private App-A or App-B using SSH ProxyCommand/ProxyJump. The private key remains on the laptop.

Application traffic: App-A → VPC A private route table → VPC peering → VPC B private route table → App-B. Return traffic uses the corresponding route from VPC B to VPC A.

Security boundary: App-A and App-B accept SSH only from the bastion's private IP (10.10.1.10/32). ICMP rules are scoped to the peer application's private IP.

3. Resource Inventory

The inventory below includes confirmed identifiers. Where an exact ID was not recorded in the lab notes, it is marked for completion from the AWS Console rather than guessed.

Resource	Name / ID	CIDR / address / notes
VPC A	ewb-vpc-a (console display was reported as “ewbvpc”) vpc-01503923cc0a241d7	10.10.0.0/16
VPC B	ewb-vpc-b vpc-0565f263f995e5199	10.20.0.0/16
Public subnet A	ewb-a-public subnet-0f6e07337186cd683	10.10.1.0/24
Private subnet A	ewb-a-private subnet-08f3d07e541c2ab30	10.10.2.0/24
Private subnet B	ewb-b-private subnet-0a7ba29df7c1a9cf8	10.20.1.0/24
Bastion EC2	i-01599364916efea24	10.10.1.10 Public IPv4: 52.66.187.199 t3.micro, Amazon Linux 2023
App-A EC2	i-0b7ac4447e487e962	10.10.2.10 No public IPv4 t3.micro, Amazon Linux 2023
App-B EC2	i-0f283a6ce2fa28c62	10.20.1.10 No public IPv4 t3.micro, Amazon Linux 2023
VPC peering	ewb-a-b-peer pcx-0a1beffef46b80661	VPC A ↔ VPC B; established and used for successful private pings
Internet Gateway	ewb-igw-a Exact ID: add from console	Attached to VPC A
Route tables	ewb-rt-a-public ewb-rt-a-private ewb-rt-b-private	Only private A route table ID recorded: rtb-04caeeb6f43586103. Add the other two IDs from console.
Security groups	ewb-sg-bastion ewb-sg-app-a ewb-sg-app-b	Exact SG IDs not recorded; see rules section.

4. IP Addressing and Subnet Design

VPC / subnet	CIDR	Purpose
VPC A	10.10.0.0/16	Public administration entry point plus a private workload subnet
ewb-a-public	10.10.1.0/24	Bastion host; route to Internet Gateway and VPC B peering
ewb-a-private	10.10.2.0/24	App-A private workload; no public IPv4
VPC B	10.20.0.0/16	Private workload VPC; no Internet Gateway
ewb-b-private	10.20.1.0/24	App-B private workload; no public IPv4

The VPC CIDRs do not overlap. Each subnet CIDR is contained within its parent VPC CIDR. This non-overlapping plan allows the VPC peering connection to route the private IPv4 ranges without address ambiguity.

5. Route Table Configuration

The three route tables have explicit subnet associations. The following routes were configured for this design; verify the current console state and attach screenshots before submission.

Route table / association	Destination	Target	Purpose
ewb-rt-a-public Associated with ewb-a-public	10.10.0.0/16	local	Within VPC A
ewb-rt-a-public	0.0.0.0/0	ewb-igw-a	Bastion internet access
ewb-rt-a-public	10.20.0.0/16	pcx-0a1beffef46b80661	Bastion-to-App-B SSH path
ewb-rt-a-private rtb-04caeeb6f43586103 Associated with ewb-a-private	10.10.0.0/16	local	Within VPC A
ewb-rt-a-private	10.20.0.0/16	pcx-0a1beffef46b80661	App-A forward route to VPC B
ewb-rt-b-private Associated with ewb-b-private	10.20.0.0/16	local	Within VPC B
ewb-rt-b-private	10.10.0.0/16	pcx-0a1beffef46b80661	Return route to VPC A

Why the return route matters

A packet can reach the destination only if the destination-side VPC has a route back to the source range. The VPC B route to 10.10.0.0/16 sends return traffic through the same peering connection. VPC peering is not transitive, and it does not allow VPC B to use VPC A's Internet Gateway as its own internet route.

6. Security Group Rules

Security group	Inbound rule	Source	Intent
ewb-sg-bastion	SSH / TCP 22	Student public IPv4 /32 at setup (reported as 103.157.160.188/32; recheck before report)	Restrict administration to the student's public IP
ewb-sg-app-a	SSH / TCP 22	10.10.1.10/32	Allow SSH only from bastion
ewb-sg-app-a	All ICMP - IPv4	10.20.1.10/32	Allow ICMP from App-B only
ewb-sg-app-b	SSH / TCP 22	10.10.1.10/32	Allow SSH only from bastion
ewb-sg-app-b	All ICMP - IPv4	10.10.2.10/32	Allow ICMP from App-A only

The default outbound allow-all rules were retained. The App-A and App-B inbound rules limit SSH and ICMP to the required lab hosts. For the controlled failure test, remove only App-B's inbound ICMP rule from 10.10.2.10/32; keep SSH in place, then restore the ICMP rule.

7. Connectivity and Validation Results

Only results actually observed during the session are recorded as passed. The baseline test was not captured before peering routes were added, so this report does not claim a baseline failure result.

Test	Command / method	Observed result	Status
Bastion SSH	Laptop → 52.66.187.199 using local ewb-lab.pem key path	Connected to Amazon Linux 2023 bastion (10.10.1.10).	PASS
Private SSH to App-A	Laptop through bastion to 10.10.2.10	Connected to Amazon Linux 2023 App-A.	PASS
Private SSH to App-B	Laptop through bastion to 10.20.1.10	Connected to Amazon Linux 2023 App-B.	PASS
Forward ICMP	On App-A: ping -c 4 10.20.1.10	4 transmitted, 4 received, 0% packet loss. Average RTT 0.280 ms.	PASS
Reverse ICMP	On App-B: ping -c 4 10.10.2.10	Four replies were observed; user reported no packet loss. Capture the complete output screenshot to preserve exact summary.	PASS*
Pre-peering baseline	On App-A before peering: ping -c 4 10.20.1.10	Not captured before routes/peering were configured; cannot be reconstructed honestly.	PENDING / NOT CAPTURED
Controlled ICMP failure	Remove App-B inbound ICMP from 10.10.2.10/32, then ping from App-A	Not yet performed in the recorded session.	PENDING
ICMP recovery	Restore App-B inbound ICMP rule and repeat forward ping	Not yet performed in the recorded session.	PENDING
Private-host isolation	EC2 details for App-A and App-B	Both were launched without public IPv4 addresses; capture the console details for evidence.	VERIFY SCREENSHOT

Forward-path explanation

App-A sends an ICMP echo request to 10.20.1.10. The private route table in VPC A matches destination 10.20.0.0/16 and forwards the packet to the peering connection. VPC B's local route delivers it to App-B, whose security group permits ICMP from 10.10.2.10/32.

Return-path explanation

App-B sends the ICMP echo reply to 10.10.2.10. VPC B's private route table matches 10.10.0.0/16 and forwards the reply through the peering connection. VPC A's local route delivers the reply to App-A. The successful reverse ping separately confirms that the opposite direction is also permitted by routes and security rules.

8. Evidence Checklist

Insert your own AWS Console and terminal screenshots in the final PDF. Crop them so the relevant resource name, status, private IP, command and output are readable. Label each screenshot with host, test and result. Do not include private keys, passwords, access tokens or other credentials.

Evidence item	What it should show	Status
E1 — VPCs and CIDRs	Both VPC IDs and CIDRs in ap-south-1	Add screenshot
E2 — Subnets	All three subnet names, CIDRs, VPCs and associations	Add screenshot
E3 — Peering	ewb-a-b-peer, Active status, both VPC IDs	Add screenshot
E4 — Route tables	All routes and subnet associations for all three tables	Add screenshot
E5 — Security groups	Inbound rules for bastion, App-A and App-B	Add screenshot
E6 — Private SSH	App-B shell reached through bastion; private IP shown	Capture/add screenshot
E7 — Forward ping	App-A command and complete 4/4 ping summary	Capture/add screenshot
E8 — Reverse ping	App-B command and complete ping summary	Capture/add screenshot
E9 — Controlled failure	Failed forward ping after removing App-B ICMP rule	Pending
E10 — Recovery	Successful forward ping after restoring App-B ICMP rule	Pending
E11 — No public IPs	EC2 instance details showing App-A and App-B without public IPv4	Add screenshot
E12 — Cleanup	Lab resources removed and final console review	Pending

9. Key Learning Outcomes

• VPC peering connects private IPv4 networks, but peering alone does not create reachability; route tables and security groups must also permit traffic.
• CIDR ranges must not overlap for this peering design.
• Both sides need routes: the source needs a route to the peer CIDR, and the peer needs a return route to the source CIDR.
• Security groups are stateful, and inbound rules can be scoped to specific peer private IPs and the bastion host.
• A bastion provides controlled administrative access to private instances without assigning them public IPv4 addresses.
• An Active peering status alone is not proof of connectivity; terminal tests and evidence are required.

10. Cleanup and Cost Control

Cleanup is not yet confirmed. After capturing all evidence, terminate the three lab EC2 instances, confirm their lab EBS volumes are removed, release any Elastic IP created, remove the lab key pair if no longer needed, delete the peering connection and peer routes, delete security groups after network interfaces disappear, remove subnets and custom route tables, detach/delete the Internet Gateway, and delete both VPCs. Finally review EC2, EBS, public IPv4 allocations, VPC resources and Billing in ap-south-1 for leftovers. Usage and billing data can take time to update.

11. Finalization Checklist

Complete these items before submitting the final version:

■ Replace the student name and batch placeholders on the cover.

■ Verify and fill in the exact Internet Gateway ID, remaining route table IDs, and security group IDs from the AWS Console.

■ Capture and insert screenshots for the resources and tests listed in Section 8.

■ Run the controlled ICMP failure test by removing only the App-B inbound ICMP rule from 10.10.2.10/32; capture the failed ping.

■ Restore the exact ICMP rule and capture a successful ping again.

■ Do not claim a baseline failure was captured before peering; if the assessor requires it, explain it was missed rather than fabricating evidence.

■ Perform cleanup after saving evidence and insert cleanup proof.

■ Export/save the final report as YourName_EWB_VPC_Peering.pdf (replace YourName with the required student name format).

12. Viva Preparation

Why is a return route needed?
Because the destination must know how to route response packets back to the source CIDR. Without it, the forward request may arrive but the reply cannot return through the intended path.

Why does App-B allow SSH from 10.10.1.10/32?
The bastion opens a separate SSH connection to App-B, and the source seen by App-B is the bastion's private IP. The laptop's public IP is used for the initial connection to the bastion.

Why doesn't Active peering guarantee a successful ping?
Routes, security group rules, network ACLs and host-level settings still determine whether the traffic is allowed and can return.

Can VPC B use VPC A's Internet Gateway through peering?
No. VPC peering does not provide transitive routing or shared internet gateway access. VPC B would need its own appropriate internet-egress design if required.

End of report draft. Add the missing evidence and complete the finalization checklist before submission.